Program 4: Maven Project To Connect To Web Servers Using Selenium Dependency (PracticeAutomation.com)

Create a Maven Project:

mvn archetype: generate -DgroupId=com.example -DartifactId=MyMavenGuavaApp -DarchetypeArtifactId=maven-archetype-quickstart -DinteractiveMode=false

```
rakshith@rakshith-VirtualBox:~/devops$ mvn archetype:generate -DgroupId=com.example -DartifactId=MyMavenSeleniumApp02 -DarchetypeArtifactId=maven-archetype-quickstart -DinteractiveMode=false
[INFO] Scanning for projects...
[INFO]
[INFO] ------------------< org.apache.maven:standalone-pom >-------------------
[INFO] Building Maven Stub Project (No POM) 1
[INFO] --------------------------------[ pom ]---------------------------------
[INFO]
[INFO] >>> maven-archetype-plugin:3.4.1:generate (default-cli) > generate-sources @ standalone-pom >>>
[INFO]
[INFO] <<< maven-archetype-plugin:3.4.1:generate (default-cli) < generate-sources @ standalone-pom <<<
[INFO]
[INFO]
[INFO] --- maven-archetype-plugin:3.4.1:generate (default-cli) @ standalone-pom ---
[INFO] Generating project in Batch mode
[INFO] No archetype defined. Using maven-archetype-quickstart (org.apache.maven.archetypes:maven-archetype-quickstart:1.0)
[INFO] ----------------------------------------------------------------------------
[INFO] Using following parameters for creating project from Old (1.x) Archetype: maven-archetype-quickstart:1.0
[INFO] ----------------------------------------------------------------------------
[INFO] Parameter: basedir, Value: /home/rakshith/devops
[INFO] Parameter: package, Value: com.example
[INFO] Parameter: groupId, Value: com.example
[INFO] Parameter: artifactId, Value: MyMavenSeleniumApp02
[INFO] Parameter: packageName, Value: com.example
[INFO] Parameter: version, Value: 1.0-SNAPSHOT
[INFO] project created from Old (1.x) Archetype in dir: /home/rakshith/devops/MyMavenSeleniumApp02
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  1.641 s
[INFO] Finished at: 2026-03-19T14:24:44+05:30
[INFO] ------------------------------------------------------------------------
```

Open pom.xml and add all the required dependencies:

```xml
<project xmlns="http://maven.apache.org/POM/4.0.0" xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
  xsi:schemaLocation="http://maven.apache.org/POM/4.0.0 http://maven.apache.org/maven-v4_0_0.xsd">
  <modelVersion>4.0.0</modelVersion>
  <groupId>com.example</groupId>
  <artifactId>MyMavenPracticeAutomationApp</artifactId>
  <packaging>jar</packaging>
  <version>1.0-SNAPSHOT</version>
  <name>MyMavenPracticeAutomationApp</name>
  <url>http://maven.apache.org</url>


  <dependencies>
    <dependency>
      <groupId>junit</groupId>
      <artifactId>junit</artifactId>
      <version>3.8.1</version>
      <scope>test</scope>
    </dependency>


    <!-- https://mvnrepository.com/artifact/org.seleniumhq.selenium/selenium-java -->
<dependency>
    <groupId>org.seleniumhq.selenium</groupId>
    <artifactId>selenium-java</artifactId>
    <version>4.29.0</version>
</dependency>
  </dependencies>
```

```
  <!-- Build: Configuring plugins and build settings -->
<build>
<plugins>
<!-- Example: Maven Compiler Plugin to compile Java code -->
<plugin>
<groupId>org.apache.maven.plugins</groupId>
<artifactId>maven-compiler-plugin</artifactId>
<version>3.8.1</version>
<configuration>
<source>1.7</source>
<target>1.7</target>
</configuration>
</plugin>
<!-- Example: Maven Surefire Plugin to run tests -->
<plugin>
<groupId>org.apache.maven.plugins</groupId>
<artifactId>maven-surefire-plugin</artifactId>
<version>2.22.2</version>
</plugin>
<plugin>
            <groupId>org.apache.maven.plugins</groupId>

            <artifactId>maven-jar-plugin</artifactId>
            <version>3.0.1</version>
           <configuration>
                <archive>
                    <manifestEntries>
                        <Main-Class>com.example.App</Main-Class>
                    </manifestEntries>
                </archive>
            </configuration>
```

```
         </plugin>


</plugins>
</build>
</project>
```

Edit App.java:

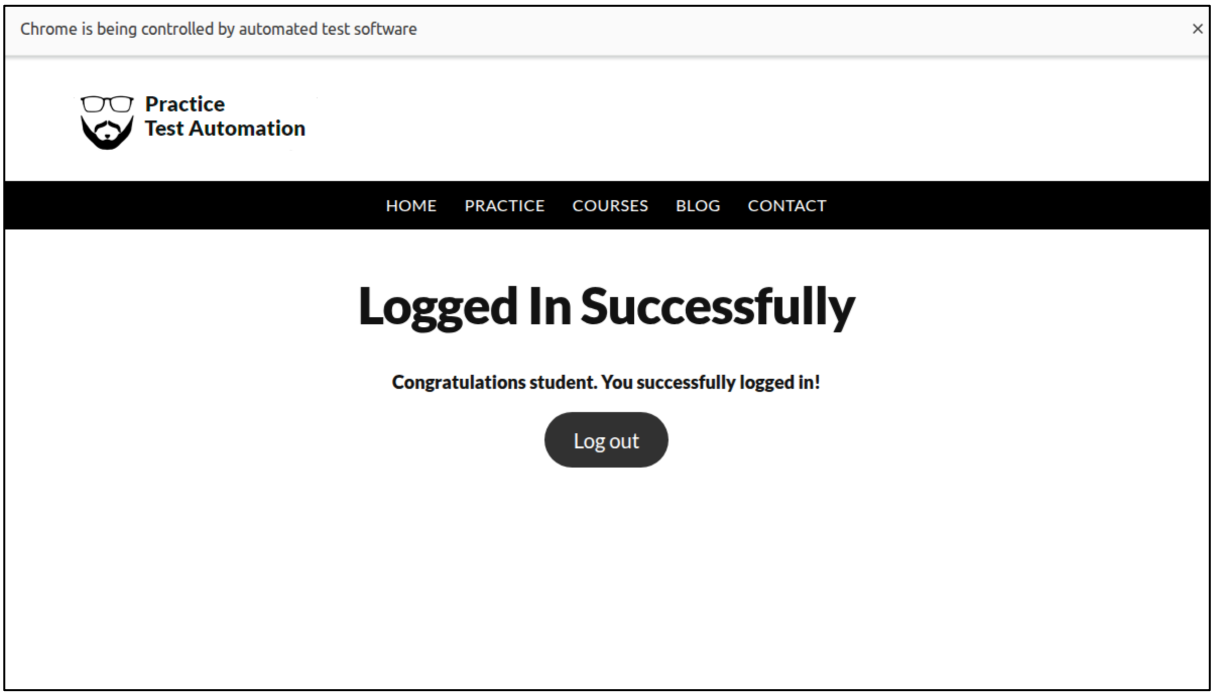
Open | App.java ~/Devops/MyMavenSeleniumApp02/src/main/java/com/example | Save

```
package com.example;
import org.openqa.selenium.By;
import org.openqa.selenium.WebDriver;
import org.openqa.selenium.chrome.ChromeDriver;
public class App
{
    public static void main( String[] args )
    {
        WebDriver driver=new ChromeDriver();
        driver.get("https://www.practicetestautomation.com/practice-test-login/");
        driver.manage().window().maximize();
        driver.findElement(By.id("username")).sendKeys("student");
        driver.findElement(By.id("password")).sendKeys("Password123");
        driver.findElement(By.id("submit")).click();
    }
}
```

Build the project using the command: mvn clean install

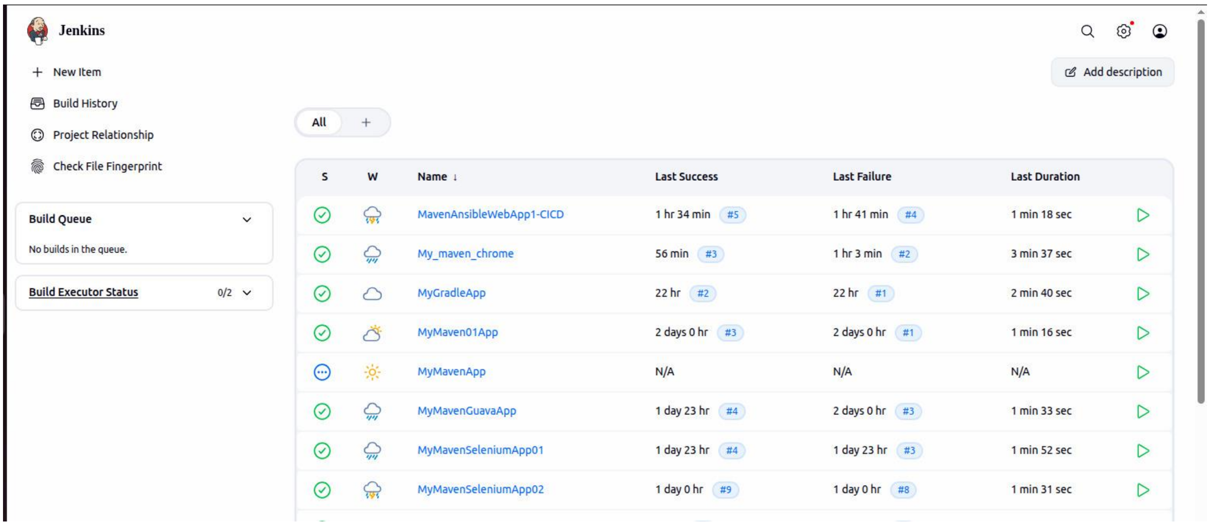
Now Run the command using

mvn exec:java -Dexec.mainClass=”com.example.App”

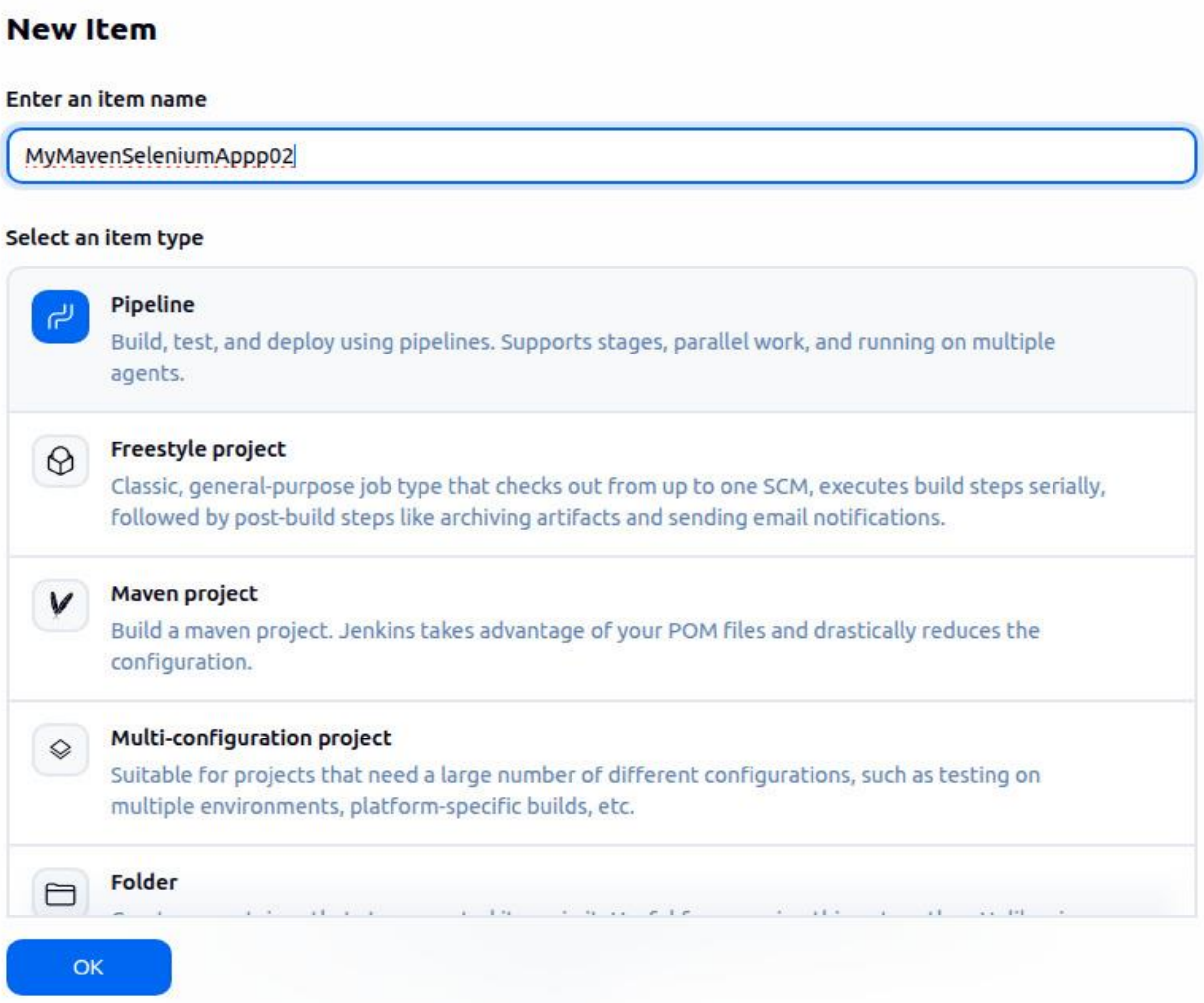


Jenkins Pipeline for MyMavenSeleniumApp02:

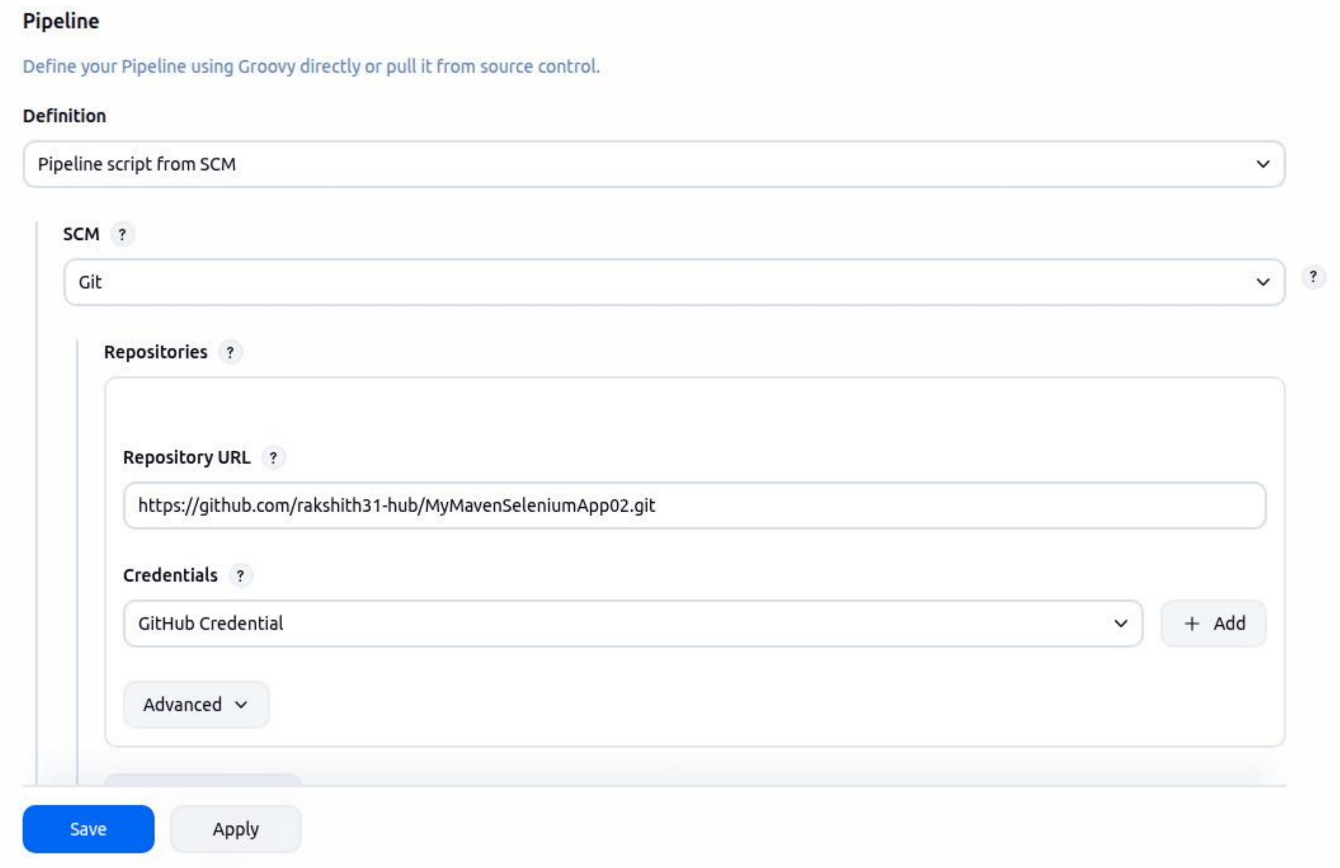
Step 1:Open Jenkins Dashboard to create a new item



<u>Step 2:</u> Click on New Item in order for creating an New Item and thus give the name as "MyMavenSeleniumApp02" and Item type as Pipeline and Create it



Step 3: Configure the settings by Defining the script as "Pipeline Script from SCM" and SCM as "Git". Thus, now as we have selected Git, we need to paste the Repository URL of the project given in GitHub and choose Credentials as "GitHub Credentials" which was created earlier.



Step 4: Now in Terminal, go to directory created for MyMavenSeleniumApp02 and create an new file called "Jenkinsfile"

Paste the code inside Jenkinsfile

Enter following code into jenkins file

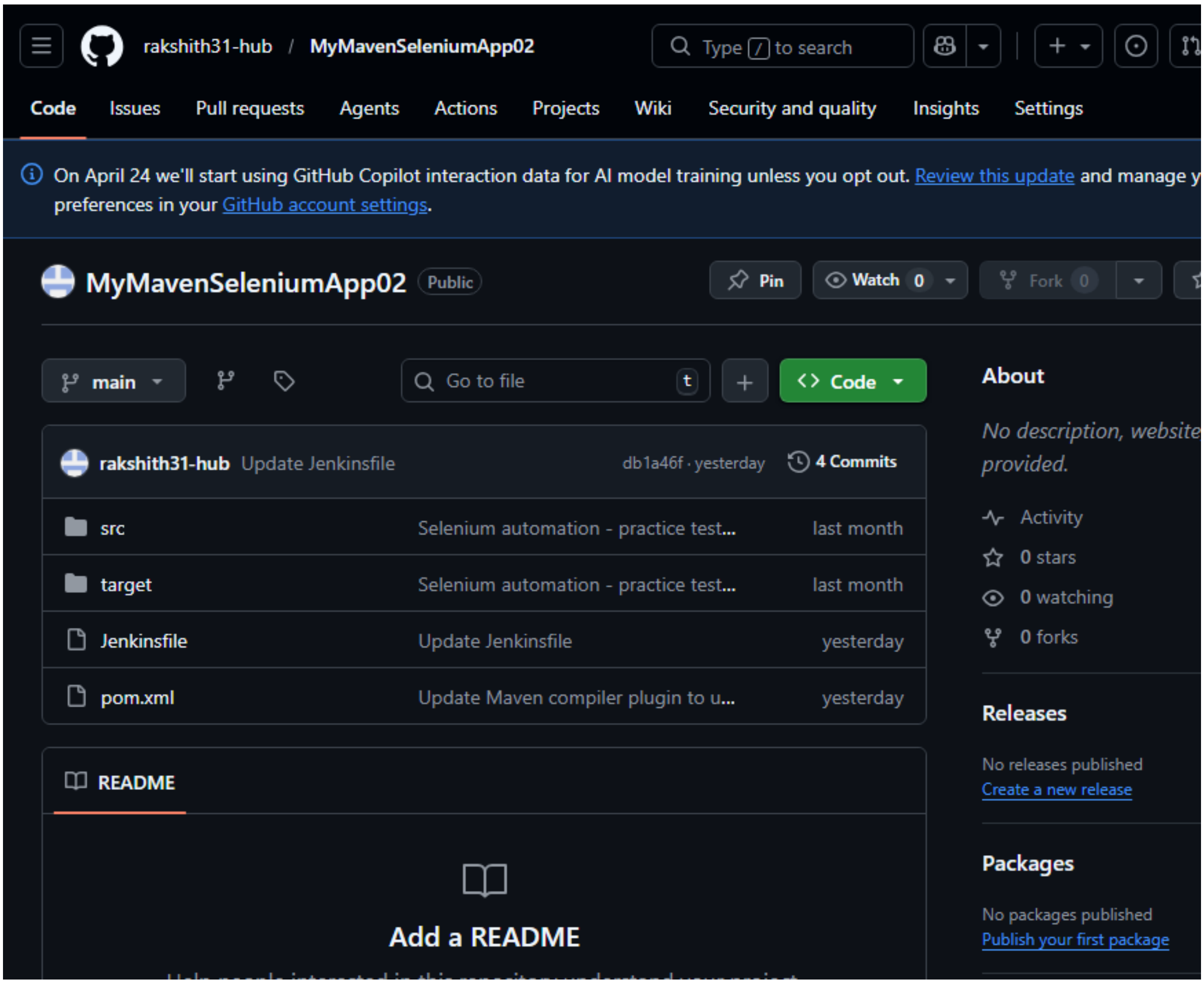
```
pipeline
{
 agent any  // Use any available agent

tools {
    maven 'Maven'  // Ensure this matches the name configured in Jenkins
}

stages {
    stage('Checkout') {
        steps {
            git branch: 'master', url:
'https://github.com/rakshith31-
hub/MymavenSeleniumApp02.git'
        }
```

```
        }


        stage('Build') { steps {
                sh 'mvn clean package'  // Run Maven build
            }

        }


        stage('Test') { steps {
                sh 'mvn test'  // Run unit tests
            }

        }




        stage('Run Application') { steps {
                // Start the JAR application
                sh 'java -jar target/MyMavenApp-1.0-SNAPSHOT.jar'
            }

        }



    }


    post {
        success {
            echo 'Build and deployment successful!'
        }

        failure {
            echo 'Build failed!'
        }

    }

}
```

<u>Step 5:</u> We need to update pom.xml also in order to change the versions of source and target from 1.7 to 1.8

```xml
<!-- Example: Maven Compiler Plugin to compile Java code -->
<plugin>
<groupId>org.apache.maven.plugins</groupId>
<artifactId>maven-compiler-plugin</artifactId>
<version>3.8.1</version>
<configuration>
<source>1.8</source>
<target>1.8</target>
</configuration>
</plugin>
```

Step 6: Now we need to Push changes and commit by pushing the necessary "Jenkinsfile" to the respective Repo



Step 7: Now go to Jenkins and Build the Pipeline and thus check the Stage view for the successful execution.